Data-Centric AI Approach for Chihuahua vs Muffin Classification

This document describes our workflow and strategy used during the 3LC x Sphere Hive AI Hackathon for the Chihuahua vs Muffin classification challenge. The competition required participants to use a fixed ResNet-18 architecture and improve model performance primarily through data-centric methods.

1. Objective

Our objective was to improve classification accuracy by systematically improving the dataset rather than modifying the model architecture. The challenge included 100 initially labeled samples and 3,579 unlabeled images. Our goal was to strategically curate and label the most informative samples.

2. Hardware and Environment

We conducted experiments using multiple machines including two GPU-enabled laptops with NVIDIA RTX 3050 GPUs and additional CPU-based systems. GPU systems were used for training iterations while CPU systems were used for data inspection, dashboard analysis, and dataset curation.

3. Data-Centric Strategy

Our methodology followed an iterative data-centric loop: Train → Analyze → Curate → Retrain.

4. Embedding Analysis

We used the 3LC dashboard to visualize feature embeddings generated by the ResNet-18 model. UMAP dimensionality reduction was applied to project high-dimensional features into 3D space. This allowed us to visually identify clusters corresponding to dogs and muffins. Cluster analysis helped us label groups of similar samples quickly and accurately.

5. Smart Sample Selection

We used several criteria to select new samples for labeling:
- High confidence predictions (>0.97) for safe dataset expansion.
- Medium confidence samples (0.75–0.95) to add diversity.
- Boundary samples (confidence 0.45–0.60) to improve decision boundaries.
- Medium loss samples (0.8–1.2) to identify potential mislabels.

6. Data Cleaning and Quality Control

We manually inspected samples with higher loss values and corrected mislabeled data when necessary. Ambiguous images were excluded from training by assigning weight = 0. This process helped remove noise and improve dataset reliability.

7. Sample Weighting Strategy

We applied 3LC sample weighting to control which samples influenced model training. Only high-confidence and carefully verified samples were given weight = 1, while uncertain or noisy samples were assigned weight = 0.

8. Iterative Training Process

Each iteration involved training the model, analyzing embeddings and metrics, selecting new samples, cleaning labels, and retraining. This iterative process was repeated multiple times to progressively improve model performance.

9. Dataset Evolution

The dataset grew from the initial 100 labeled images to over 700 curated samples. The final dataset maintained a balanced distribution between dog and muffin classes to prevent bias during training.

10. Results

Through our data-centric workflow, embedding analysis, and dataset refinement, we achieved a validation accuracy exceeding 93%. The improvement came primarily from dataset quality improvements rather than architectural changes.

11. Key Insights

- Data quality can outperform complex model changes.
- Embedding visualization is extremely powerful for identifying clusters.
- Small corrections to mislabeled samples significantly improve accuracy.
- Boundary samples help models learn difficult distinctions.

12. Conclusion

This project demonstrates the effectiveness of a data-centric AI approach. By focusing on data quality, embedding-driven labeling, and strategic sample selection, we significantly improved model performance while keeping the architecture fixed.